

# Spot an Error

## And WIN Award!

You can win one thousand rupees plus a certificate from the editor of Electronics For You magazine if you spot and report any factual error in the latest issue of the magazine. Your report will be published in a coming issue of the magazine, giving credit to you.

Though every care is taken in preparation of the content for the magazine, an error can slip in at times. So we invite our learned readers to point out any error that they spot, for the benefit of the other readers.

### Rules:

- The error has to be reported before end of the month that is mentioned on cover of the magazine
- The report has to be sent in writing or by email addressed to the Editor, EFY Magazine ([editsec@efyindia.com](mailto:editsec@efyindia.com))
- The error should be of factual nature, in the information published in the magazine, and not a grammatical or syntax error
- The correct information should also be provided along with the error spotted by you
- In case more than one person reports the same error, the one whose report reaches us the earliest will get the award money
- In case two or more persons jointly report an error, the award money will be divided amongst them
- EFY employees, their family members and friends, and EFY associates and their family members and friends are not eligible
- Decision of the editor will be final in this matter

**electronics**  
FOR YOU

**Q1.** Is there any method to convert pressure generated from human walking into electricity using piezoelectric transducers? I have seen an article indicating that this can be achieved. Kindly guide me how it is possible and what components, material and circuitry are needed to build this project to get a detectable output.

Abhishek

Through e-mail

**A1.** Piezoelectric generators work due to the piezoelectric effect. This is the ability of certain materials to create electrical potential when responding to mechanical changes. In simple words, when compressed, expanded or otherwise changing shape, a

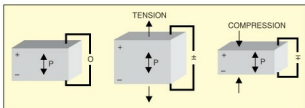


Fig. 1: Piezoelectric effect—the ability of certain materials to create electrical potential when responding to mechanical changes



Fig. 2: Piezoelectric power from shoes

piezoelectric material will output some voltage. This effect is also possible in reverse in the sense that putting a charge through the material will result in changing shape or mechanical stress.

Piezoelectric materials are useful in a variety of ways. Certain piezoelectric materials can handle high voltage extremely well and are useful in transformers and other electrical components. These are also used to make motors and reduce vibration in sensitive environments. They can also be used as an energy collector.

The piezoelectric effect is found in a number of natural and man-made materials. Commonly used naturally-occurring crystals include quartz, to-

paz, tourmaline, Rochelle salts and cane sugar. Man-made crystals include the quartz-like langasite and gallium orthophosphate. Common piezoelectric man-made ceramics include barium titanate, lead titanate and lead zirconate titanate. Other naturally-occurring piezoelectric materials include dry bone, tendons, silk, some woods, enamel, dentin and collagen.

The electrical charge created by piezo crystals is fairly low. To compensate for this, these are usually used in high-repetition applications. People move thousands of times each day. Piezo crystals can be embedded in everyday apparel like shoes. Every time a person takes a step, the crystal generates a small charge. Over time, with thousands of steps, these small charges build up until the amount becomes significant. This energy can be used to keep personal electronics, such as cell phones and MP3 players, fully charged.

You can follow below-mentioned steps to make a piezoelectric generator:

1. Install under the shoe multiple piezoelectric transducers that you can find easily in the market.
2. Connect the leads from piezoelectric transducers to capacitors or a diode bridge. Such bridges smooth out the energy collected from the piezoelectric transducers by regulating voltage, so a constant voltage passes downstream to batteries or the load.
3. Connect the leads from the capacitors or diode bridge to batteries for storage. This provides a constant source of electricity for your load, be it a street sign, a street light or a cellphone charging station.

**Q2.** In India, why is standard mains voltage 220V/230V, 50Hz—not 110V/115V, 60Hz as in some foreign countries?

Ramakanta Mohanta

Through e-mail

**A2.** The choice of mains distribution voltage is mainly influenced by